Appendix. Spectral Vegetation Indices Implemented in the Hyperspectral Wheat-Image Processing Pipeline

Scope and radiometric basis

This appendix specifies the 37 spectral indices implemented in the project. Rλ denotes the calibrated signal or relative reflectance at wavelength λ (nm), according to the selected computational mode. The pipeline optionally subtracts a dark frame and applies geometric and spatial corrections. Spectralon reflectance is assumed to equal 1 at all measured wavelengths for normalization. The indices differ in their sensitivity to chlorophyll, carotenoids, anthocyanins, water absorption, leaf structure, canopy background and photoprotection; none is intrinsically the best index for all physiological questions.

Spectral sampling and interpolation

Acquisitions may use wavelength increments of 5, 10, 15 or 20 nm. For each required center λ, the corresponding measured band is used directly if present. Otherwise, a missing center within the acquired spectral range is reconstructed by linear interpolation between the neighboring measured wavelengths: $R(\lambda) = R(\lambda1) + [R(\lambda2) - R(\lambda1)]\ (\lambda - \lambda1)/(\lambda2 - \lambda1)$. If any required wavelength lies outside the acquired range, the index is recorded as NA in the sample table and a same-size blank white PNG is generated for its map.

Implemented Indices

NDVI — Normalized Difference Vegetation Index

$$(R800 - R680)/(R800 + R680)$$

Required bands (nm): 800, 680

Distinctive value and preferred use: A useful baseline for identifying green vegetation, estimating relative canopy cover and comparing gross greenness across wheat samples. Its red band approaches strong chlorophyll absorption, while 800 nm responds to internal scattering and leaf structure. Unlike chlorophyll-focused red-edge indices, NDVI also responds to leaf area, exposed background and shadows. In dense, strongly green tissue it can saturate, so a small change in chlorophyll may produce little NDVI change. Use NDVI as a general reference against which more specific indices are evaluated, not as a stand-alone chlorophyll assay.

Reference: Tucker C.J. Red and photographic infrared linear combinations for monitoring vegetation. Remote Sens. Environ. 1979, 8, 127–150. https://doi.org/10.1016/0034-4257(79)90013-0

NDVI_670 — Normalized Difference Vegetation Index, 670 nm variant

$$(R800 - R670)/(R800 + R670)$$

Required bands (nm): 800, 670

Distinctive value and preferred use: This variant shifts the red band from 680 to 670 nm, closer to a strong chlorophyll-a absorption region. It may respond differently to moderate chlorophyll changes and permits sensitivity testing when both bands are measured. Because the two indices share their 800-nm channel and the same algebra, they are largely redundant rather than independent physiological traits.

Prefer this variant when 670 nm offers a stronger and cleaner plant signal; the optimal band depends on spectral resolution, noise and the sample's pigment level. Like conventional NDVI, it can saturate in dark-green material.

Reference: Tucker C.J. Red and photographic infrared linear combinations for monitoring vegetation. Remote Sens. Environ. 1979, 8, 127–150. https://doi.org/10.1016/0034-4257(79)90013-0

GNDVI — Green Normalized Difference Vegetation Index

$$(R800 - R550)/(R800 + R550)$$

Required bands (nm): 800, 550

Distinctive value and preferred use: Replacing the red band of NDVI with 550 nm moves sensitivity toward the green reflectance peak, which can remain responsive to pigment changes when red absorption is already strong. It may therefore be informative for comparing chlorophyll-related variation or nitrogen-associated greenness in otherwise similar green wheat leaves. The trade-off is pronounced sensitivity to pigments other than chlorophyll, particularly anthocyanins that darken the green region. For purple/red foliage, a red-edge index is generally a more defensible chlorophyll candidate; GNDVI should instead be treated as a combined pigment-color signal.

Reference: Gitelson A.A.; Kaufman Y.J.; Merzlyak M.N. Use of a green channel in remote sensing of global vegetation from EOS-MODIS. Remote Sens. Environ. 1996, 58, 289–298. https://doi.org/10.1016/S0034-4257(96)00072-7

NDRE — Normalized Difference Red Edge

$$(R790 - R720)/(R790 + R720)$$

Required bands (nm): 790, 720

Distinctive value and preferred use: The 720-nm channel samples the lower red-edge slope, while 790 nm lies in the near-infrared shoulder. This emphasizes chlorophyll-dependent spectral-shape changes that can persist after red-based NDVI has approached saturation. It is particularly attractive for dark-green wheat, subtle chlorophyll loss and comparisons among similarly developed leaves. In anthocyanin-rich leaves, the red-edge region generally avoids much of the green-band interference, although pigment overlap and leaf structure still matter. Relative to NDVI705, this version uses a more distant NIR reference and a higher red-edge wavelength; neither configuration is universally superior.

Reference: Gitelson A.A.; Merzlyak M.N. Signature analysis of leaf reflectance spectra: algorithm development for remote sensing of chlorophyll. J. Plant Physiol. 1996, 148, 494–500. https://doi.org/10.1016/S0176-1617(96)80284-7

NDVI705 — Red-edge NDVI, 750/705

$$(R750 - R705)/(R750 + R705)$$

Required bands (nm): 750, 705

Distinctive value and preferred use: The 705-nm band lies near the steep red-edge transition, and 750 nm provides an upper-edge reference. This shorter-baseline normalized difference can respond

markedly to local changes in red-edge slope and hence to chlorophyll variation within individual leaves. Compared with NDRE (790/720), it uses both a lower measuring band and a closer reference band; results may diverge when red-edge position or leaf scattering changes. It is useful for detailed pigment comparisons, including strongly colored material where a green-band index is confounded, but requires careful wavelength registration and interpolation because the slope is steep near 705 nm.

Reference: Gitelson A.A.; Merzlyak M.N. Signature analysis of leaf reflectance spectra: algorithm development for remote sensing of chlorophyll. J. Plant Physiol. 1996, 148, 494–500. https://doi.org/10.1016/S0176-1617(96)80284-7

CIgreen — Green Chlorophyll Index

$$R790/R550-1$$

Required bands (nm): 790, 550

Distinctive value and preferred use: Unlike the bounded normalized GNDVI, CIgreen uses the ratio of 790-nm to 550-nm reflectance minus one. This can provide a broader numerical response to chlorophyll-related changes when the green denominator decreases, making it useful for calibration against measured leaf chlorophyll in uniformly green plants. However, the ratio can be strongly amplified by low green reflectance, noise or anthocyanin absorption. CIgreen and GNDVI largely probe the same spectral pair, so reporting both does not add an independent pigment measurement. Prefer CIrededge for chlorophyll work on anthocyanin-rich foliage.

Reference: Gitelson A.A. et al. Remote estimation of canopy chlorophyll content in crops. Geophys. Res. Lett. 2005, 32, L08403. https://doi.org/10.1029/2005GL022688

CIrededge — Red-edge Chlorophyll Index

$$R790/R710-1$$

Required bands (nm): 790, 710

Distinctive value and preferred use: This ratio replaces CIgreen's pigment-sensitive green band with 710 nm, near the lower red edge. It is advantageous when high chlorophyll makes visible-red indices insensitive or when anthocyanins strongly alter the green part of the spectrum. Published pigment-retrieval work specifically recommends red-edge, rather than green, bands for chlorophyll estimation in anthocyanin-containing leaves. Compared with NDRE or NDVI705, CIrededge is an unbounded ratio and may offer a wide dynamic range but is more sensitive to a small or noisy denominator. Use with leaf-level calibration; it does not automatically eliminate structural effects.

Reference: Gitelson A.A. et al. Remote estimation of canopy chlorophyll content in crops. Geophys. Res. Lett. 2005, 32, L08403. https://doi.org/10.1029/2005GL022688

Additional reference: Gitelson, A.A.; Keydan, G.P.; Merzlyak, M.N. Three-band model for noninvasive estimation of chlorophyll, carotenoids, and anthocyanin contents in higher plant leaves. Geophys. Res. Lett. 2006, 33, L11402. https://doi.org/10.1029/2006GL026457

SR — Simple Ratio Vegetation Index

$$R800/R680$$

Required bands (nm): 800, 680

Distinctive value and preferred use: SR is the simple 800/680 near-infrared-to-red ratio, often responsive to green biomass or leaf-area variation. It differs from NDVI only mathematically: with identical bands, SR = (1 + NDVI)/(1 - NDVI), so the two carry the same ideal spectral information. The unbounded SR scale can spread apart high-greenness observations, but it also amplifies errors when red reflectance is very small. Retain SR if a calibration model or published comparison requires a ratio scale; otherwise NDVI is easier to compare and interpret. Neither uniquely distinguishes chlorophyll from geometry or biomass.

Reference: Jordan C.F. Derivation of leaf-area index from quality of light on the forest floor. Ecology 1969, 50, 663–666. https://doi.org/10.2307/1936256

SR705 — Simple Ratio Red Edge, 750/710

$$R750/R710$$

Required bands (nm): 750, 710

Distinctive value and preferred use: This upper-to-lower red-edge ratio focuses on chlorophyll-linked spectral contrast at 750 and 710 nm instead of the conventional NIR/red contrast used in SR. It can be useful when highly chlorophyllous tissue causes the visible-red denominator of SR to become saturated or very small. Relative to NDVI705 it uses 710 rather than 705 nm and removes normalized-difference bounding; it may better spread high-contrast values but become unstable when the 710-nm channel is weak. Best used on reliably aligned, calibrated spectra with adequate signal-to-noise around the red edge.

Reference: Gitelson A.A.; Merzlyak M.N. Signature analysis of leaf reflectance spectra: algorithm development for remote sensing of chlorophyll. J. Plant Physiol. 1996, 148, 494–500. https://doi.org/10.1016/S0176-1617(96)80284-7

VOG1 — Vogelmann Red-edge Index 1

$$R740/R720$$

Required bands (nm): 740, 720

Distinctive value and preferred use: VOG1 compares 740 with 720 nm, targeting a narrow section of the upper red-edge slope rather than the wider red-to-NIR contrast captured by NDRE. It can reveal subtle red-edge shifts related to pigment and leaf condition, particularly among samples with similar total brightness. Its narrow wavelength separation also makes it more susceptible to sensor-band placement, spectral interpolation and local noise. Prefer broader red-edge indices when wavelength spacing is coarse or the 720/740-nm measurements are unstable; use VOG1 for higher-resolution investigations of local red-edge shape.

Reference: Vogelmann J.E.; Rock B.N.; Moss D.M. Red edge spectral measurements from sugar maple leaves. Int. J. Remote Sens. 1993, 14, 1563–1575. https://doi.org/10.1080/01431169308953986

MTCI — MERIS Terrestrial Chlorophyll Index

$$(R750 - R710)/(R710 - R680)$$

Required bands (nm): 750, 710, 680

Distinctive value and preferred use: MTCI estimates the relative slopes of two adjacent red-edge segments, (750-710) and (710-680) nm, rather than relying on a single reflectance ratio. It is useful when chlorophyll changes redistribute the shape of the red-edge transition and the objective is to compare chlorophyll-related variation across relatively developed canopies. Compared with CIrededge, it emphasizes slope geometry, but it can become extremely unstable if the lower segment's reflectance difference approaches zero. Its exact 680/710/750-nm configuration is a project band adaptation; evaluate denominator stability and interpolate cautiously.

Reference: Dash J.; Curran P.J. The MERIS terrestrial chlorophyll index. Int. J. Remote Sens. 2004, 25, 5403–5413. https://doi.org/10.1080/0143116042000274015

PRI — Photochemical Reflectance Index

$$(R531 - R570)/(R531 + R570)$$

Required bands (nm): 531, 570

Distinctive value and preferred use: PRI targets the reflectance feature near 531 nm associated with xanthophyll-cycle photoprotection, using 570 nm as a reference. It is therefore complementary to NDVI and chlorophyll indices: under controlled short-term measurements it may indicate changes in photosynthetic light-use efficiency even before visible greenness changes. Across longer periods it also reflects pigment-pool composition, canopy geometry and illumination, so it is not a direct photosynthetic-rate measurement. Choose PRI for controlled, comparable lighting and sufficiently narrow spectral bands. If 531 nm is interpolated from 5-20-nm-spaced bands, the narrow underlying feature cannot be fully reconstructed.

Reference: Gamon J.A.; Peñuelas J.; Field C.B. A narrow-waveband spectral index that tracks diurnal changes in photosynthetic efficiency. Remote Sens. Environ. 1992, 41, 35–44. https://doi.org/10.1016/0034-4257(92)90059-S

PRI570_530 — Photochemical Reflectance Index, 530 nm approximation

$$(R530 - R570)/(R530 + R570)$$

Required bands (nm): 530, 570

Distinctive value and preferred use: This deliberately approximate PRI replaces the canonical 531-nm measuring center with 530 nm, which may coincide with an acquired camera channel and avoid interpolation. For 10- or 20-nm sampling, 530 nm may be operationally more stable than a numerically interpolated 531-nm value, but the two should not automatically be regarded as identical: a narrow xanthophyll-related feature is being sampled at different centers. Report this variant separately rather than merging values with PRI across experiments. It serves the same physiological question as PRI and is not an independent indicator of stress.

Reference: Gamon J.A.; Peñuelas J.; Field C.B. A narrow-waveband spectral index that tracks diurnal changes in photosynthetic efficiency. Remote Sens. Environ. 1992, 41, 35–44. https://doi.org/10.1016/0034-4257(92)90059-S

PSRI — Plant Senescence Reflectance Index

$$(R680 - R500)/R750$$

Required bands (nm): 680, 500, 750

Distinctive value and preferred use: PSRI emphasizes changes in the relative balance of chlorophyll and persistent accessory pigments during senescence. Unlike NDVI, which describes overall green-versus-NIR contrast, PSRI uses visible bands at 500 and 680 nm normalized by 750 nm; it may reveal pigment breakdown before substantial loss of leaf area. It is especially relevant to leaf aging, yellowing and prolonged stress. It is less specific for an immediate, reversible stress response than PRI, and pigment composition, anthocyanins and illumination can still influence its value. Interpret higher or lower values against a consistent developmental baseline.

Reference: Sims D.A.; Gamon J.A. Relationships between leaf pigment content and spectral reflectance across a wide range of species, leaf structures and developmental stages. Remote Sens. Environ. 2002, 81, 337–354. https://doi.org/10.1016/S0034-4257(02)00010-X

SIPI — Structure Insensitive Pigment Index

$$(R800 - R445)/(R800 - R680)$$

Required bands (nm): 800, 445, 680

Distinctive value and preferred use: SIPI is designed to follow carotenoid-to-chlorophyll changes while reducing some influence of leaf structural scattering through the 800-nm reference. In comparison with PSRI, it uses a blue band near 445 nm and may distinguish pigment-composition shifts that occur without marked senescence. This makes it useful for studying photoprotective pigments, maturation or stress-induced chlorophyll loss. Nevertheless, the blue band is sensitive to scattering and other absorbers, and the denominator (R800-R680) can become small in atypical or poorly calibrated spectra. SIPI is a relative pigment proxy, not a direct carotenoid concentration.

Reference: Sims D.A.; Gamon J.A. Relationships between leaf pigment content and spectral reflectance across a wide range of species, leaf structures and developmental stages. Remote Sens. Environ. 2002, 81, 337–354. https://doi.org/10.1016/S0034-4257(02)00010-X

ARI1 — Anthocyanin Reflectance Index 1

$$1/R550 - 1/R700$$

Required bands (nm): 550, 700

Distinctive value and preferred use: ARI1 exploits the reciprocal reflectance contrast at 550 and 700 nm to track anthocyanin accumulation. It is particularly useful for distinguishing red/purple pigmentation from ordinary chlorophyll-related greenness: a leaf may contain abundant chlorophyll yet have strongly altered green reflectance because of anthocyanins. Unlike CIgreen, this index is intended

to describe that anthocyanin interference rather than to ignore it. The reciprocal form depends strongly on reflectance scale and on a low-noise green band; apply it to calibrated reflectance and consider chlorophyll background effects.

Reference: Gitelson A.A.; Merzlyak M.N.; Chivkunova O.B. Optical properties and nondestructive estimation of anthocyanin content in plant leaves. Photochem. Photobiol. 2001, 74, 38–45. https://doi.org/10.1562/0031-8655(2001)074%3C0038:OPANEO%3E2.0.CO;2

ARI2 — Anthocyanin Reflectance Index 2

$$R800 * (1/R550 - 1/R700)$$

Required bands (nm): 550, 700, 800

Distinctive value and preferred use: ARI2 multiplies the ARI1 anthocyanin contrast by 800-nm reflectance. This adds a near-infrared structural reference that can help account for differences in leaf optical path and scattering when comparing anthocyanin-rich leaves. Use it alongside ARI1 to examine whether apparent anthocyanin differences depend on structural variation or total NIR brightness. Both indices remain susceptible to chlorophyll and leaf anatomy; the NIR weighting does not make ARI2 a universally superior anthocyanin assay. Its relative scaling behavior also differs from unweighted ARI1.

Reference: Gitelson A.A.; Merzlyak M.N.; Chivkunova O.B. Optical properties and nondestructive estimation of anthocyanin content in plant leaves. Photochem. Photobiol. 2001, 74, 38–45. https://doi.org/10.1562/0031-8655(2001)074%3C0038:OPANEO%3E2.0.CO;2

CRI1 — Carotenoid Reflectance Index 1

$$1/R510 - 1/R550$$

Required bands (nm): 510, 550

Distinctive value and preferred use: CRI1 contrasts reciprocal reflectance at 510 and 550 nm, exploiting carotenoid absorption in the blue-green region. It is relevant when carotenoid variation is the main question, for example during photoacclimation or pigment turnover, rather than overall canopy greenness. Compared with CRI2 it uses the green 550-nm reference, which may work well in anthocyanin-free leaves but can be severely confounded in red/purple leaves. Chlorophyll absorption and the reciprocal calculation also influence the result; use calibrated reflectance and, ideally, independent pigment measurements.

Reference: Gitelson A.A.; Zur Y.; Chivkunova O.B.; Merzlyak M.N. Assessing carotenoid content in plant leaves with reflectance spectroscopy. Photochem. Photobiol. 2002, 75, 272–281. https://pubmed.ncbi.nlm.nih.gov/11950093/

CRI2 — Carotenoid Reflectance Index 2

$$1/R510 - 1/R700$$

Required bands (nm): 510, 700

Distinctive value and preferred use: CRI2 compares 510 nm with a 700-nm reference instead of 550 nm. This may reduce interference by anthocyanins that strongly absorb in the green region, making it

preferable to CRI1 when leaves are visibly red or purple. The compromise is that 700 nm lies close to the chlorophyll-sensitive red edge, so variable chlorophyll content can introduce another source of interference. The 510-nm band still reflects multiple pigments. CRI2 is best interpreted as a carotenoid-related candidate within a pigment-panel analysis, not as a selective direct measurement.

Reference: Gitelson A.A.; Zur Y.; Chivkunova O.B.; Merzlyak M.N. Assessing carotenoid content in plant leaves with reflectance spectroscopy. Photochem. Photobiol. 2002, 75, 272–281. https://pubmed.ncbi.nlm.nih.gov/11950093/

WBI — Water Band Index (900/970)

$$R900/R970$$

Required bands (nm): 900, 970

Distinctive value and preferred use: WBI probes the weak water-absorption region around 970 nm relative to 900 nm, providing a water-status dimension largely absent from NDVI, CIrededge and other chlorophyll-centered indices. In the implemented direction R900/R970, deeper absorption at 970 nm can increase the ratio. It is useful for assessing relative dehydration when camera sensitivity and calibration near 970 nm are adequate. Interpretation is affected by leaf thickness, path length, NIR scattering and instrumental noise at the edge of the measured range. Check that the chosen ratio direction matches literature comparisons.

Reference: Peñuelas J. et al. The reflectance at the 950–970 nm region as an indicator of plant water status. Int. J. Remote Sens. 1993, 14, 1887–1905. https://doi.org/10.1080/01431169308954010

NRI — Normalized Red/Green Nitrogen-related Index (570/670)

$$(R570-R670)/(R570+R670)$$

Required bands (nm): 570, 670

Distinctive value and preferred use: This normalized 570/670-nm green-versus-red contrast was included as a nitrogen-related optical proxy. It can track changes in crop nitrogen status insofar as nitrogen availability changes chlorophyll and associated pigmentation. Unlike NDRE or CIrededge, however, it contains no NIR or red-edge reference and is easily influenced by illumination, senescence and non-nitrogen stress. It should therefore be treated as a locally calibrated empirical indicator rather than a nitrogen-specific spectral absorption index. Its sign and magnitude may not be comparable with indices elsewhere labelled 'NRI' but using different bands.

Reference: Xue J.; Su B. Significant Remote Sensing Vegetation Indices: A Review of Developments and Applications. J. Sens. 2017, 2017, 1353691. https://doi.org/10.1155/2017/1353691

RGI — Red–Green Index (690/550)

$$R690/R550$$

Required bands (nm): 690, 550

Distinctive value and preferred use: The 690/550-nm ratio compares deep-red and green reflectance and can reveal changes in leaf coloration, pigment degradation and visible-band contrast. It is more

color-oriented than NDVI or NIR-based chlorophyll ratios, potentially useful for distinguishing visibly yellowing or reddish leaves even when their NIR scattering remains similar. However, 690 nm is still chlorophyll sensitive and 550 nm is affected by anthocyanins, so the index cannot separate chlorophyll loss from red pigmentation on its own. Prefer it for exploratory visible-pigment comparisons with ARI and PSRI rather than as a single stress diagnostic.

Reference: Sims D.A.; Gamon J.A. Relationships between leaf pigment content and spectral reflectance across a wide range of species, leaf structures and developmental stages. Remote Sens. Environ. 2002, 81, 337–354. https://doi.org/10.1016/S0034-4257(02)00010-X

R750_R700 — Red-edge Chlorophyll Ratio (750/700)

$$R750 / R700$$

Required bands (nm): 750, 700

Distinctive value and preferred use: This simple red-edge ratio spans 700 to 750 nm, capturing a chlorophyll-sensitive lower-edge channel and an upper-edge reference. Relative to CIrededge (790/710), it moves the sensitive channel closer to the visible-red transition and shortens the spectral baseline, which may heighten sensitivity to changes in red-edge position but also to wavelength error. It is a compact indicator for monitoring chlorophyll-related differences in leaves with comparable structure. When the lower channel has very low reflectance or strong interpolation uncertainty, a normalized red-edge difference may be more stable.

Reference: Gitelson A.A.; Merzlyak M.N. Signature analysis of leaf reflectance spectra: algorithm development for remote sensing of chlorophyll. J. Plant Physiol. 1996, 148, 494–500. https://doi.org/10.1016/S0176-1617(96)80284-7

mND705 — Modified Normalized Difference 705

$$(R750 - R705) / (R750 + R705 - 2 * R445)$$

Required bands (nm): 750, 705, 445

Distinctive value and preferred use: The modified ND705 includes a 445-nm term in addition to the 705/750-nm red-edge pair. This adjustment attempts to exploit additional pigment-related information beyond a two-band red-edge index and can alter discrimination when leaf pigment composition varies. It is worth comparing directly with NDVI705, rather than assuming it is always superior. The extra blue band is affected by carotenoids, chlorophyll and scattering; coarse spectral sampling or low blue-band signal can add substantial noise. Do not interpret a gain in numerical separation as greater chlorophyll specificity without validation.

Reference: Sims D.A.; Gamon J.A. Relationships between leaf pigment content and spectral reflectance across a wide range of species, leaf structures and developmental stages. Remote Sens. Environ. 2002, 81, 337–354. https://doi.org/10.1016/S0034-4257(02)00010-X

MSR705 — Modified Simple Ratio 705

$$(R750 - R445)/(R705 - R445)$$

Required bands (nm): 750, 445, 705

Distinctive value and preferred use: MSR705 forms a ratio of the 750- and 705-nm signals after subtracting the 445-nm band from each, in contrast to the blue-adjusted normalized difference mND705. It may accentuate pigment-driven differences that are compressed in conventional SR705 or NDVI705, but the denominator (R705-R445) can approach zero, yielding extremely unstable values. Use it on well-calibrated, high-quality leaf spectra when the aim is to test whether blue-band adjustment improves chlorophyll-related prediction. It is not necessarily preferable with coarse band spacing or noisy blue wavelengths.

Reference: Sims D.A.; Gamon J.A. Relationships between leaf pigment content and spectral reflectance across a wide range of species, leaf structures and developmental stages. Remote Sens. Environ. 2002, 81, 337–354. https://doi.org/10.1016/S0034-4257(02)00010-X

REP — Red Edge Position (four-band interpolation)

$$700 + 40 * ((R670 + R780)/2 - R700)/(R740 - R700)$$

Required bands (nm): 670, 780, 700, 740

Distinctive value and preferred use: REP summarizes the estimated wavelength of the red-edge transition, expressed in nanometers rather than as a dimensionless index. This makes it useful for investigating whether treatment, chlorophyll content or stress changes the location of the steep spectral slope, a distinction not apparent from NDVI alone. In many green leaves increasing chlorophyll is associated with a shift to longer wavelengths, but water status, leaf structure and pigment mixtures also contribute. The four-band interpolation becomes unstable when the 700-to-740-nm slope is weak; REP should be examined with its denominator and spectral sampling in mind.

Reference: Vogelmann J.E.; Rock B.N.; Moss D.M. Red edge spectral measurements from sugar maple leaves. Int. J. Remote Sens. 1993, 14, 1563–1575. https://doi.org/10.1080/01431169308953986

MCARI — Modified Chlorophyll Absorption in Reflectance Index

$$((R700 - R670) - 0.2 * (R700 - R550)) * (R700/R670)$$

Required bands (nm): 700, 670, 550

Distinctive value and preferred use: MCARI estimates a chlorophyll-related red absorption feature while partially compensating for the green-to-red spectral baseline. Compared with NDVI, it concentrates more directly on the local absorption shape rather than on broad vegetation cover. It can be valuable for quantitative chlorophyll calibration when leaf structure or canopy coverage varies, although it is still affected by soil background and canopy architecture. The index is scale-dependent; it should be computed from relative reflectance rather than uncalibrated digital numbers. MCARI/OSAVI is preferable to MCARI alone when exposed background is a major source of variation.

Reference: Haboudane D. et al. Integrated narrow-band vegetation indices for prediction of crop chlorophyll content for application to precision agriculture. Remote Sens. Environ. 2002, 81, 416–426. https://doi.org/10.1016/S0034-4257(02)00018-4

TCARI — Transformed Chlorophyll Absorption in Reflectance Index

$$3*((R700-R670)-0.2*(R700-R550)*(R700/R670))$$

Required bands (nm): 700, 670, 550

Distinctive value and preferred use: TCARI is a transformed chlorophyll-absorption metric intended to emphasize absorption behavior while modifying the baseline correction found in MCARI. It offers an alternative response scale for chlorophyll calibration and may prove useful when MCARI remains correlated with background or structural differences. However, the implemented placement of the R700/R670 factor differs from some published TCARI expressions; results must be identified as this project's formula and not automatically compared numerically with standard TCARI studies. Like MCARI, it requires calibrated reflectance and does not intrinsically remove soil effects; the TCARI/OSAVI combination is designed for that problem.

Reference: Haboudane D. et al. Integrated narrow-band vegetation indices for prediction of crop chlorophyll content for application to precision agriculture. Remote Sens. Environ. 2002, 81, 416–426. https://doi.org/10.1016/S0034-4257(02)00018-4

TVI — Triangular Vegetation Index

$$0.5*(120*(R750-R550)-200*(R670-R550))$$

Required bands (nm): 750, 550, 670

Distinctive value and preferred use: The Triangular Vegetation Index describes the approximate area of a spectral triangle spanning the green, red and near-infrared regions. Unlike NDVI's bounded two-band contrast, TVI incorporates three spectral points and can respond to the shape of chlorophyll absorption and the green reflectance peak. It may therefore help separate spectral-shape differences among leaves with broadly similar NDVI. Its absolute magnitude also responds to canopy brightness, background and calibration. Use it as a structural/pigment-related complement rather than a direct estimate of photosynthetic rate; here TVI does not mean the Transformed Vegetation Index.

Reference: Xue J.; Su B. Significant Remote Sensing Vegetation Indices: A Review of Developments and Applications. J. Sens. 2017, 2017, 1353691. https://doi.org/10.1155/2017/1353691

DVI — Difference Vegetation Index

$$R800-R680$$

Required bands (nm): 800, 680

Distinctive value and preferred use: DVI is the simple NIR-minus-red difference. It measures the absolute spectral separation between structural scattering and red absorption, whereas NDVI divides the same difference by total brightness. DVI may preserve useful amplitude information when samples share a consistent radiometric scale, but it is particularly sensitive to exposure, illumination and leaf

thickness. If such factors vary, NDVI usually provides more robust comparisons. DVI and NDVI use the same two channels, so they should be treated as complementary mathematical scalings of the same spectral contrast, not independent physiological measurements.

Reference: Tucker C.J. Red and photographic infrared linear combinations for monitoring vegetation. Remote Sens. Environ. 1979, 8, 127–150. https://doi.org/10.1016/0034-4257(79)90013-0

RDVI — Renormalized Difference Vegetation Index

$$(R800 - R670)/\sqrt{(R800 + R670)}$$

Required bands (nm): 800, 670

Distinctive value and preferred use: RDVI partially renormalizes the NIR-minus-red difference by the square root of total reflectance. This places it between amplitude-sensitive DVI and strongly normalized NDVI, a useful compromise when vegetation cover, soil exposure and overall brightness vary moderately. It may preserve sensitivity where NDVI approaches saturation without being as dependent on absolute brightness as DVI. It is not, however, a full background correction and is scale-dependent. Prefer RDVI for calibrated canopy or mixed plant-background data, and compare it with SAVI/OSAVI if exposed soil is substantial.

Reference: Roujean J.-L.; Breon F.-M. Estimating PAR absorbed by vegetation from bidirectional reflectance measurements. Remote Sens. Environ. 1995, 51, 375–384. https://doi.org/10.1016/0034-4257(94)00114-3

OSAVI — Optimized Soil Adjusted Vegetation Index

$$1.16 * (R800 - R670)/(R800 + R670 + 0.16)$$

Required bands (nm): 800, 670

Distinctive value and preferred use: OSAVI adds a fixed 0.16 soil-adjustment term to a red/NIR normalized difference, specifically targeting variation introduced by exposed background. Relative to SAVI with L = 0.5, its smaller correction often retains more vegetation contrast while reducing moderate soil effects. It is most useful for partially covered plots, seedlings or canopy imagery with visible substrate. For pixel masks that already exclude nearly all background, its advantage over NDVI may be small. Because of the additive constant, OSAVI requires reflectance-scaled data; it must not be interpreted from arbitrary brightness-scaled previews.

Reference: Rondeaux G.; Steven M.; Baret F. Optimization of soil-adjusted vegetation indices. Remote Sens. Environ. 1996, 55, 95–107. https://doi.org/10.1016/0034-4257(95)00186-7

SAVI — Soil Adjusted Vegetation Index, L=0.5

$$1.5 * (R800 - R680)/(R800 + R680 + 0.5)$$

Required bands (nm): 800, 680

Distinctive value and preferred use: SAVI uses an explicit soil-adjustment parameter L = 0.5 to moderate the sensitivity of red/NIR contrast to background brightness. It is particularly suitable for early-stage crops and sparse vegetation where the soil occupies a substantial part of the sensor's field of

view. Compared with OSAVI, the larger adjustment provides a different compromise between soil suppression and vegetation responsiveness; neither is consistently better across all cover fractions. SAVI is unnecessary as a universal replacement for NDVI in tightly cropped leaf masks. The L term only has its intended meaning on a reflectance scale.

Reference: Huete A.R. A soil-adjusted vegetation index (SAVI). Remote Sens. Environ. 1988, 25, 295–309. https://doi.org/10.1016/0034-4257(88)90106-X

EVI — Enhanced Vegetation Index

$$2.5*(R800-R680)/(R800+6*R680-7.5*R450+1)$$

Required bands (nm): 800, 680, 450

Distinctive value and preferred use: EVI combines an enhanced red/NIR contrast with a 450-nm correction, originally introduced to improve vegetation sensitivity in densely covered scenes while addressing atmospheric/background influences. It can be informative when NDVI saturates in very green canopies. In close-range laboratory wheat imaging, however, atmospheric aerosol correction is often irrelevant, while the blue band and additive constants may amplify calibration errors. Use EVI only on correctly scaled reflectance with reliable blue-channel signal; it is not automatically superior to NDRE or CIrededge for leaf chlorophyll estimation.

Reference: Huete A. et al. Overview of the radiometric and biophysical performance of the MODIS vegetation indices. Remote Sens. Environ. 2002, 83, 195–213. https://doi.org/10.1016/S0034-4257(02)00096-2

MSAVI — Modified Soil Adjusted Vegetation Index

$$(2*R800+1-\sqrt{((2*R800+1)^2-8*(R800-R670)))}/2$$

Required bands (nm): 800, 670

Distinctive value and preferred use: MSAVI applies a nonlinear soil-adjustment scheme rather than SAVI's fixed L = 0.5. It is intended to adapt more flexibly across vegetation cover conditions and can be useful when comparing wheat seedlings with widely varying fractions of visible soil. Compared with OSAVI/SAVI, it requires no user-chosen soil parameter, but its nonlinear expression can be unstable or uninterpretable on uncalibrated DN values. For isolated masked leaves, the background advantage is limited and chlorophyll-specific red-edge indices may be more informative. Check the radicand and validity of all pixel-level values.

Reference: Qi J. et al. A modified soil adjusted vegetation index. Remote Sens. Environ. 1994, 48, 119–126. https://doi.org/10.1016/0034-4257(94)90134-1

MCARI_OSAVI — MCARI / OSAVI ratio

$$MCARI/OSAVI$$

Required bands (nm): 700, 670, 550, 800

Distinctive value and preferred use: Dividing MCARI by OSAVI combines a chlorophyll-absorption measure with a soil-adjusted vegetation contrast. This can reduce variation associated with canopy cover

or background when comparing chlorophyll-related signals across plots, potentially outperforming MCARI alone in incomplete canopies. Its advantage is much less certain for fully segmented single leaves, where exposed soil has already been excluded. The ratio is unstable when OSAVI is near zero and the index remains sensitive to the calibration scale. Treat it as a model candidate for chlorophyll retrieval rather than a direct, universally transferable concentration measure.

Reference: Haboudane D. et al. Integrated narrow-band vegetation indices for prediction of crop chlorophyll content for application to precision agriculture. Remote Sens. Environ. 2002, 81, 416–426. https://doi.org/10.1016/S0034-4257(02)00018-4

TCARI_OSAVI — TCARI / OSAVI ratio

$$TCARI / OSAVI$$

Required bands (nm): 700, 670, 550, 800

Distinctive value and preferred use: TCARI/OSAVI similarly normalizes a transformed chlorophyll-absorption signal by a soil-adjusted index and is especially relevant when canopy chlorophyll must be estimated in the presence of variable cover or soil brightness. Compared with MCARI/OSAVI, it places more weight on the transformed absorption construction and may show different sensitivity to pigment, background or noise. This project's TCARI formula is not identical to every published TCARI definition, so its ratio cannot be equated uncritically with literature values. As with MCARI/OSAVI, denominator stability and reflectance calibration are essential.

Reference: Haboudane D. et al. Integrated narrow-band vegetation indices for prediction of crop chlorophyll content for application to precision agriculture. Remote Sens. Environ. 2002, 81, 416–426. https://doi.org/10.1016/S0034-4257(02)00018-4